

TOPS Technologies

Security and Maintenance

Assessment A3 — Complete Answers

Theme: Small Business IT Setup

Total Time: 3 Hours

Assessment 3

Section A: Scenario-Based Concept Application

1. How the installation of Active Directory Domain Services (AD DS)

AD DS centralizes authentication and authorization, stores users, groups, and computers in one database, enables Single Sign-On, and simplifies administration and security.

2.Explain how GPOs enforce password complexity and account lockout

GPOs apply password complexity, minimum length, password history, expiration, and account lockout policies consistently across the domain.

3.Explain NTFS and Share permissions.

Share permissions control network access, NTFS controls file/folder access, and the most restrictive effective permission is applied.

4.Explain DHCP DORA.

Discover, Offer, Request, Acknowledge automatically assigns unique IP addresses and avoids conflicts.

5.Explain DNS Name Resolution.

DNS translates hostnames into IP addresses so users can access resources using names.

6.Explain Windows Server Backup Schedule.

Configure scheduled backups using Windows Server Backup Wizard for automatic daily or weekly backups.

Section B: Practical Application

Task 1 Core Services:

Step 1: Install Windows Server 2022

- Create a new virtual machine in VMware/VirtualBox.
- Mount the Windows Server 2022 ISO.
- Install Windows Server (Desktop Experience).
- Set the Administrator password and log in.

Step 2: Configure a Static IP Address

- Open Control Panel
 - Network and Sharing Center

- Change adapter settings.
- Right-click Ethernet
 - Properties.
- Select Internet Protocol Version 4 (TCP/IPv4).
- Assign:
 - IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1 Preferred DNS Server: 192.168.1.10

Step 3: Install AD DS and DNS Roles

- Open Server Manager.
- Click Manage
 - Add Roles and Features.
- Select Role-based or feature-based installation.
- Select the local server.
- Check:
 - Active Directory Domain Services (AD DS) DNS Server
- Click Next
 - Install.
- Wait until the installation completes.

Step 4: Promote the Server to a Domain Controller

- In Server Manager, click the notification flag.
- Select Promote this server to a domain controller.
- Choose Add a new forest.
- Enter the Root Domain Name, for example:
 - Technova.local
- Set the Directory Services Restore Mode (DSRM) password.
- Keep the default settings and click Next until the Install button appears.
- Click Install.
- The server will restart automatically.

Step 5: Verify Installation

- Log in using the domain Administrator account.
- Open Server Manager and confirm that:
 - Active Directory Domain Services is installed.
 - DNS Server is installed.
- Open Active Directory Users and Computers to verify the domain.
- Open DNS Manager to verify that the DNS zone has been created.

Task 2 Directory Structure:

Step 1:Open Active Directory Administrative Center

- Log in to the Domain Controller.
- Open Server Manager.
- Click Tools
 - Active Directory Administrative Center (ADAC)

Step 2:Create 3 Organizational Units (OUs)

- Expand your domain (e.g., technova.local).
- Right-click the domain
 - New
 - Organizational Unit.
- Create the following OUs:
 - HR
 - Sales
 - IT

Step 3:Create 5 User Accounts

- Open the required OU.
- Right-click
 - New
 - User.
- Enter:
 - Full Name
 - User Logon Name
 - Password
- Select User must change password at next logon (optional).
- Click OK.

Step 4:Create 2 Security Groups

- Right-click the domain or desired OU.
- Select New
 - Group.
- Enter the group name.
- Choose:
 - Group Scope: Global
 - Group Type: Security
- Click OK.

Steps 5: Add Users to Security Groups

Assign users as follows:

- HR_Group
 - hruser1
 - hruser2
- Sales_Group
 - salesuser1
 - salesuser2

Task 3 Policy Enforcement:

Step 1 Open Group Policy Management

- Log in to the Domain Controller.
- Open Server Manager
 - Tools
 - Group Policy Management.

Step 2 Create a New GPO

- Expand your domain (e.g., technova.local).
- Right-click the required OU (e.g., Sales)
 - Create a GPO in this domain, and Link it here.
- Name the GPO Security Policy.

Step 3 Configure Password Policy

- Right-click the GPO
 - Edit.
- Go to:
 - Computer Configuration
 - Policies
 - Windows Settings
 - Security Settings
 - Account Policies
 - Password Policy
- Configure:
 - Enforce password history: 5 passwords
 - Maximum password age: 30 days
 - Minimum password length: 8 characters
 - Password must meet complexity requirements: Enabled

Step 4 Restrict Control Panel

- In the same GPO, navigate to:
 - User Configuration
 - Policies
 - Administrative Templates
 - Control Panel
- Open Prohibit access to Control Panel and PC settings.
- Select Enabled
 - Apply
 - OK.

Step 5 Apply the Policy

- Run:
 - gpupdate /force

Task 4 Resource Management:

Step 1 Create a Folder

- Create folder
 - D/ ComanyData

Step 2 Enable Sharing

- Right-click the folder → Properties.
- Open the Sharing tab.
- Click Advanced Sharing.
- Check Share this folder.
- Share Name:
 - CompanyData

Step 3 Configure Share Permissions

- HR_Group
 - Change
- Sales_Group
 - Read
- Administrators
 - Full Control

Step 4 Configure NTFS Permissions

- Open the Security tab.
- Remove unnecessary users if required.
- Add:

- HR_Group
 - Modify
- Sales_Group
 - Read & Execute
- Administrators
 - Full Control

Step 5 Test Access

- Log in as an HR user and verify read/write access.
- Log in as a Sales user and verify read-only access.

Task 5 Network Automation:

Step 1 Install DHCP Server

- Open Server Manager.
- Select Manage → Add Roles and Features.
- Install the DHCP Server role.

Step 2 Authorize DHCP

- Open DHCP Manager.
- Right-click the server.
- Click Authorize.
- Refresh until the server shows a green icon.

Step 3 Create an IPv4 Scope

- Expand IPv4.
- Select New Scope.
- Configure:
 - Scope Name: OfficeScope
 - Start IP: 192.168.1.100
 - End IP: 192.168.1.200
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
 - DNS Server: 192.168.1.10
- Activate the scope.

Step 4 Verify on Client

- Open Command Prompt and run:
 - ipconfig /release
 - ipconfig /renew

- ipconfig /all

Section C: Integrated Case Study

1:Analysis

Deny NTFS permissions, incorrect GPO security filtering, GPO replication failure.

2:Troubleshooting.

Use rsop.msc or gpresult /h report.html, gpupdate /force, Event Viewer.

3:CLI/GUI tools.

gpresult, whoami /groups, icacls, net share, Group Policy Management, ADUC.

4:Resolution.

Correct permissions, replicate GPOs, use AGDLP.

5:Validation.

Use net share and icacls to verify share and NTFS permissions.